ADI Digital Design Summer Internship 2026

Layered TB for 8-bit ALU

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ANALOG DEVICES

AHEAD OF WHAT'S POSSIBLE™

1. Transcript

Transcript

```
# [TEST] Starting test with 100 transactions
# [GEN] Starting generation of 100 transactions
# [GEN] Done generating 100 transactions
# [SB] PASS: reset -> alu_out == 0
# [SB] PASS: reset -> alu_out == 0
# [SB] PASS: reset -> alu_out == 0
# [DRV] Reset complete
# [SB] PASS: rst=1 fun=SUB en=0 a=a b=4b | out=be valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=ad b=4 | out=a4 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=f6 b=a8 | out=ad valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=71 b=dd | out=fe valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=50 b=8e | out=a1 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=0 a=a b=b5 | out=0 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=c9 b=29 | out=1e valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=e0 b=a | out=90 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=76 b=2b | out=b9 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=b3 b=30 | out=77 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=57 b=4e | out=7f valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=c8 b=86 | out=72 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=7d b=b2 | out=80 valid=1
# [SB] PASS: rst=1 fun=SUB en=0 a=c5 b=ea | out=ff valid=1
# [SB] PASS: rst=1 fun=SUB en=1 a=2c b=59 | out=dd valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=29 b=59 | out=d3 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=31 b=cd | out=9 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=3b b=7 | out=84 valid=1
# [SB] PASS: rst=1 fun=SUB en=0 a=72 b=94 | out=98 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=11 b=fb | out=0 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=90 b=93 | out=4e valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=33 b=96 | out=90 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=aa b=28 | out=b7 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=1 b=36 | out=56 valid=1
# [SB] PASS: rst=1 fun=SUB en=1 a=31 b=e1 | out=0 valid=1
# [SB] PASS: rst=1 fun=ADD en=0 a=65 b=b2 | out=50 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=25 b=19 | out=ed valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=e9 b=e2 | out=fd valid=1
# [SB] PASS: rst=1 fun=ADD en=0 a=fc b=7a | out=eb valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=56 b=63 | out=bd valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=ba b=91 | out=77 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=99 b=2b | out=90 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=4b b=b4 | out=bb valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=29 b=29 | out=45 valid=1
# [SB] PASS: rst=1 fun=ADD en=0 a=dc b=d1 | out=29 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=0 a=dc b=99 | out=4 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=0 a=b0 b=31 | out=18 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=fc b=32 | out=6d valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=ec b=5f | out=30 valid=1
# [SB] PASS: rst=1 fun=ADD en=1 a=e6 b=8a | out=4c valid=1
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# [SB] PASS: rst=1 fun=OR_OP en=1 a=e9 b=e2 | out=fd valid=1
# [SB] PASS: rst=1 fun=ADD en=0 a=fc b=7a | out=eb valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=56 b=63 | out=bd valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=ba b=91 | out=77 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=99 b=2b | out=90 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=4b b=b4 | out=bb valid=1
# [SB] PASS: rst=1 fun=OR_OP en=1 a=29 b=29 | out=45 valid=1
# [SB] PASS: rst=1 fun=ADD en=0 a=dc b=d1 | out=29 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=0 a=dc b=99 | out=4 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=0 a=b0 b=31 | out=18 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=fc b=32 | out=6d valid=1
# [SB] PASS: rst=1 fun=AND_OP en=1 a=ec b=5f | out=30 valid=1
# [SB] PASS: rst=1 fun=ADD en=1 a=e6 b=8a | out=4c valid=1
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# [SB] PASS: rst=1 fun=AND_OP en=0 a=90 b=98 | out=70 valid=1
# [SB] PASS: rst=1 fun=ADD en=0 a=f9 b=7b | out=12 valid=1
# [SB] PASS: rst=1 fun=OR_OP en=0 a=b8 b=98 | out=24 valid=1
# [SB] PASS: rst=1 fun=AND_OP en=0 a=30 b=3 | out=18 valid=1
# [MON] Done monitoring 102 samples
# ====================================================
# [SB] Scoreboard summary: matches=44 mismatches=0
# ====================================================
# [DRV] Done driving 100 transactions
# [TEST] Test complete
# [TB] Simulation finished
# ** Note: $finish    : alu_tb.sv(40)
```

2. Waveform snippet